🤖 INSTRUCTOR-LED COURSE SYLLABUS · PYTHON MACHINE LEARNING

Machine Learning Foundations

An instructor-led course in 11 three-hour sessions. Learn supervised machine learning the way it is done at work: frame the question, prepare the data without leaking, train and compare models fairly, tune them, and report results you can defend, all practiced on one fictional company's data.

Course site: rays-python-machine-learning.netlify.app

`11 sessions × 3 h` `33 class hours` `21 lessons` `2–3 h homework per session` `Python 3.12+`

📚 COURSE DESCRIPTION

Machine learning is mostly not about clever algorithms. It is about asking a precise question, preparing data honestly, comparing models fairly against a baseline, and knowing when a score is too good to be true. This course teaches those habits first and then the main families of models, with scikit-learn, in the order you would use them on a real project.

Every session uses the same company: **Pinecrest Outfitters**, a fictional outdoor-gear retailer with an online shop and four stores. Its seven data files were built for machine learning: customers who might churn, weekly product demand, card payments with rare fraud, product reviews, purchase histories and daily store sales, each with real patterns to find and a few traps (a leaking column, a rare class, a drifting rate) to catch.

🎯 By the end you will be able to

- Frame a business request as a classification or regression problem with a clear target
- Split data honestly (stratified, grouped, by time), catch target leakage, and cross-validate
- Choose the right metric, and set a decision threshold from the costs of mistakes
- Impute, scale, encode and engineer features inside leak-proof pipelines
- Handle rare classes with class weights, resampling and thresholds
- Train and read linear models, k-nearest neighbors, naive Bayes, SVMs, trees, forests and gradient boosting
- Compare models fairly, tune them with cross-validation and Optuna, and report the result honestly

Continue with **Advanced Machine Learning** (13 sessions): explaining models, clustering and anomaly detection, dimensionality reduction, deep learning with PyTorch, text and recommendations, MLOps and a capstone.

✅ PREREQUISITES

- Python and pandas at the level of **Python Data Science** (free at Ray's House of Fun): writing functions, filtering and grouping DataFrames, handling missing values, and making a basic Matplotlib chart.
- No machine-learning experience is needed. Comfortable reading a formula with a sum in it; no calculus.
- A Windows, macOS or Linux laptop on which you can install software, with about 3 GB free. Setup is done at home before Session 1 (it follows Lesson 0.1).

💡 FORMAT

11 sessions × 3 hours = 33 class hours, two lessons per session (the studio session 11 focuses on one). Every lesson on the course site is written for a solo learner and takes about 90 to 120 minutes, so two lessons do not fit whole in one room session. The room protects what is hardest alone: the **mental model**, a **live-coding demo** you type along with, a **guided lab** with help on hand, and **starting** each lesson project. At home you finish the projects and do each lesson's exercises, quiz and journal entry on the free course site: about **two to three hours** between sessions.

Each session runs 0:00 to 3:00: a warm-up on last session, the first lesson, a 10-minute break, the second lesson, and a wrap-up with an exit ticket and the next bring list.

Twice weekly: about 6 weeks. Two homework sets a week, so plan on four to six hours outside class.

Weekly: 11 weeks, with a full week for each homework set.

Intensive days: two sessions a day; homework becomes evening practice and the final project is finished afterward.

📖 SESSION BY SESSION

Lesson titles link to the lesson pages on the course site. Homework is the time to finish that session's lesson projects, exercises, quizzes and journal entries.

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
1 **Your ML Toolkit and the Estimator API** WORKFLOW	0.1 Your ML Toolkit 1.1 The scikit-learn Estimator API	Set up a reproducible machine-learning environment, read a dataset like a data scientist (one row, the target, the gaps and why), and train a baseline, a logistic regression and a random forest with scikit-learn's shared fit/predict/score API.	2–3 h
2 **Splits, Leakage and Cross-Validation** WORKFLOW	1.2 Splits, Baselines and Leakage 1.3 Cross-Validation	Hold out test data properly (stratified, by time, with a validation set), catch a leaking column by its too-good-to-be-true score, and use k-fold, grouped and time-series cross-validation, reading the spread as well as the mean.	2–3 h
3 **Metrics and Preprocessing** WORKFLOW PREPARING DATA	1.4 Metrics for Classification and Regression 2.1 Impute, Scale and Encode	Judge models with precision, recall, ROC and PR curves, MAE, RMSE and R^2, choose a threshold from the costs of mistakes, and impute, scale and encode mixed columns with ColumnTransformer.	2–3 h
4 **Pipelines and Creating Features** PREPARING DATA	2.2 Pipelines 2.3 Creating Features	Chain preprocessing and a model into one leak-proof pipeline you can cross-validate and save, and create date, ratio, aggregate and interaction features that give a model something to learn.	2–3 h

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
5 **Feature Selection and Imbalanced Classes** PREPARING DATA	2.4 Feature Selection 2.5 Imbalanced Classes	Drop redundant and useless features with filter, wrapper and embedded methods, and model a rare event (fraud) with class weights, resampling and a deliberately chosen threshold.	2–3 h
6 **Linear Models and Regularization** LINEAR MODELS	3.1 Linear and Logistic Regression 3.2 Regularization and the Bias-Variance Trade-off	Fit, read and check linear and logistic regression, and control overfitting with Ridge, Lasso and Elastic Net, explaining the bias-variance trade-off. Includes Checkpoint 1.	2–3 h
7 **Nearest Neighbors and Naive Bayes** CLASSIC CLASSIFIERS	4.1 k-Nearest Neighbors 4.2 Naive Bayes	Classify by similarity with k-nearest neighbors (and see why scaling and k matter), and build a fast probabilistic classifier with naive Bayes, including a first text model.	2–3 h
8 **Support Vector Machines and Decision Trees** CLASSIC CLASSIFIERS TREES & ENSEMBLES	4.3 Support Vector Machines 5.1 Decision Trees	Train maximum-margin classifiers with linear and kernel SVMs, and grow, draw, read and prune decision trees, watching how they overfit.	2–3 h
9 **Random Forests and Gradient Boosting** TREES & ENSEMBLES	5.2 Bagging and Random Forests 5.3 Gradient Boosting	Build random forests with out-of-bag error, and train gradient-boosted trees with HistGradientBoosting, XGBoost and LightGBM, using early stopping.	2–3 h
10 **Ensembles and Model Selection** TREES & ENSEMBLES SELECTION & TUNING FINAL PROJECT	5.4 Voting and Stacking 6.1 Model Selection and Learning Curves	Combine models by voting and stacking, compare candidates on equal terms with learning and validation curves, and choose a final project with a one-paragraph plan.	2–3 h
11 **Hyperparameter Tuning and Final Project Studio** SELECTION & TUNING FINAL PROJECT	6.2 Hyperparameter Tuning	Tune models with grid search, random search, successive halving and Optuna inside nested cross-validation, then carry a supervised project from question to an honest, reproducible result and present it.	2–3 h

Total out-of-class time: about 22 hours or more across the course.

🕑 SESSION OUTLINES

Times run from the start of each session (0:00 to 3:00). Shift the block to any start time.

SESSION 1 Your ML Toolkit and the Estimator API

You build: 0.1 A data card for the seven course files (data_overview.csv) and your learning journal; **1.1** A churn-risk shortlist: the 50 customers most likely to leave, checked against what really happened

0:00–0:30	Welcome, environment check and diagnostic
0:30–1:25	`0.1` Your ML Toolkit frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`1.1` The scikit-learn Estimator API frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 2 Splits, Leakage and Cross-Validation

You build: 1.2 An honest evaluation of a churn model: a leak audit, a validation-based choice and one final test score, saved as a short report; **1.3** A fair bake-off of four churn models on repeated folds, with one final test-set check

0:00–0:10	Warm-up
0:10–1:25	`1.2` Splits, Baselines and Leakage frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`1.3` Cross-Validation frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 3 Metrics and Preprocessing

You build: 1.4 A retention-campaign plan: the threshold that maximizes expected profit, checked once on the test set, plus a call list for a team with limited capacity; **2.1** A model-ready version of the full churn table, and churn scores for brand-new customers with gaps and an unfamiliar store

0:00–0:10	Warm-up
0:10–1:25	`1.4` Metrics for Classification and Regression frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`2.1` Impute, Scale and Encode frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 4 Pipelines and Creating Features

You build: 2.2 A deployable churn pipeline: chosen by cross-validation, tested once, saved to a file and reloaded to score raw new customers; **2.3** A weekly demand forecaster whose engineered features cut the error of a linear model by about 30%

0:00–0:10	Warm-up
0:10–1:25	2.2 Pipelines frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	2.3 Creating Features frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 5 Feature Selection and Imbalanced Classes

You build: 2.4 A lean churn model that uses 6 of 16 columns and scores as well as the full model; **2.5** A fraud review queue for June, trained on January to May, with a threshold chosen to minimize the cost of fraud plus reviews

0:00–0:10	Warm-up
0:10–1:25	2.4 Feature Selection frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	2.5 Imbalanced Classes frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 6 Linear Models and Regularization

You build: 3.1 A price-response model for Pinecrest's weekly demand that estimates how much a discount really sells, and forecasts 2025 better than a naive baseline; **3.2** A weekly demand forecast with season-by-category interactions, regularized by cross-validation and tested on 2025

> Checkpoint 1 (30 minutes): a short open-notebook task on Modules 1 and 2.

0:00–0:10	Warm-up
0:10–0:40	**Checkpoint 1 (Modules 1–2)**
0:40–1:40	`3.1` Linear and Logistic Regression frame and mental model, live demo, guided lab, start the lesson project
1:40–1:50	*Break*
1:50–2:50	`3.2` Regularization and the Bias-Variance Trade-off frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 7 Nearest Neighbors and Naive Bayes

You build: 4.1 A lookalike churn model whose every prediction comes with the real customers that justify it; **4.2** A review-triage tool that ranks reviews for the customer-care team and flags star ratings that disagree with the text

0:00–0:10	Warm-up
0:10–1:25	`4.1` k-Nearest Neighbors frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`4.2` Naive Bayes frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 8 Support Vector Machines and Decision Trees

You build: 4.3 A tuned SVM retention list for Pinecrest, compared fairly with logistic regression; **5.1** A churn rule book: a small tree's rules and its customer segments ranked by churn rate, checked against a baseline and logistic regression

0:00–0:10	Warm-up
0:10–1:25	`4.3` Support Vector Machines frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`5.1` Decision Trees frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 9 Random Forests and Gradient Boosting

You build: 5.2 A churn forest in a pipeline, validated out-of-bag, compared with Lesson 5.1's rule tree and logistic regression, plus a 50-customer call list and an importance check; **5.3** A weekly demand forecaster for 24 products in 4 stores, trained on 2024 and tested on 2025

0:00–0:10	Warm-up
0:10–1:25	5.2 Bagging and Random Forests frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	5.3 Gradient Boosting frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 10 Ensembles and Model Selection

You build: 5.4 A churn-model bake-off: three single models against a voting and a stacking ensemble, with a ship-or-skip recommendation backed by a bootstrap; **6.1** A churn-model bake-off: five candidates scored on identical folds, a learning curve for the winner, one final test score and a results file

> Final project kickoff: choose churn, demand or fraud and leave with a written plan.

0:00–0:10	Warm-up
0:10–1:15	5.4 Voting and Stacking frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:30	6.1 Model Selection and Learning Curves frame and mental model, live demo, guided lab, start the lesson project
2:30–2:50	**Final project: choose and plan**
2:50–3:00	Wrap-up

SESSION 11

Hyperparameter Tuning and Final Project Studio

You build: 6.2 A tuned gradient-boosting churn model for the retention shortlist, compared honestly with the default, with its settings saved to best_params.json

> Studio with milestone check-ins, then a showcase of every learner's project.

0:00–0:10	Warm-up
0:10–1:10	6.2 Hyperparameter Tuning frame and mental model, live demo, guided lab
1:10–1:20	*Break*
1:20–2:15	**Final project studio**
2:15–2:50	**Showcase**
2:50–3:00	Wrap-up

📅 SAMPLE SCHEDULES

No dates: fit them to your calendar. "Twice weekly" puts two sessions in each week (Day 1 and Day 2), finishing in week 6.

SESSION	TWICE WEEKLY (≈6 WEEKS)	WEEKLY (11 WEEKS)
1 Your ML Toolkit and the Estimator API	Week 1, Day 1	Week 1
2 Splits, Leakage and Cross-Validation	Week 1, Day 2	Week 2
3 Metrics and Preprocessing	Week 2, Day 1	Week 3
4 Pipelines and Creating Features	Week 2, Day 2	Week 4
5 Feature Selection and Imbalanced Classes	Week 3, Day 1	Week 5
6 Linear Models and Regularization	Week 3, Day 2	Week 6
7 Nearest Neighbors and Naive Bayes	Week 4, Day 1	Week 7
8 Support Vector Machines and Decision Trees	Week 4, Day 2	Week 8
9 Random Forests and Gradient Boosting	Week 5, Day 1	Week 9
10 Ensembles and Model Selection	Week 5, Day 2	Week 10
11 Hyperparameter Tuning and Final Project Studio	Week 6, Day 1	Week 11

🏁 CHECKPOINT AND PROJECT

- **Self-checks and exit tickets** in every session are practice, not grades: they tell you and your instructor what to revisit.
- **Checkpoint 1 (Session 6, 30 minutes):** an individual, open-notebook task on the workflow and data preparation (Modules 1 and 2), scored with a short rubric.

- **Final project.** In Session 10 you choose one Pinecrest problem (who will churn, how many units each store will sell, or which payments are fraud) and write a one-paragraph plan. Session 11 is a studio with milestone check-ins, then a showcase. You hand in a notebook that reruns from top to bottom and a short report: the framing, an honest split and baseline, a tuned model compared fairly with simpler ones, the metric that matters to the business, and what you would and would not claim. A well-supported modest result scores as well as a dramatic one.

🧰 REQUIRED SOFTWARE AND DATA

- **Python 3.12 or newer:** with a virtual environment for the course
- **The course packages:** one command: `python -m pip install -r requirements.txt` (scikit-learn 1.9, pandas 3.0, NumPy, Matplotlib, XGBoost, LightGBM, imbalanced-learn, Optuna, and the rest of the course's list, including PyTorch for later modules)
- **JupyterLab or VS Code:** JupyterLab installs with the packages; VS Code needs its Python and Jupyter extensions
- **The course data:** `pinecrest-ml-data.zip` (1.4 MB) from the course home page
- **No GPU, accounts or paid tools:** everything runs on an ordinary laptop processor

Everything installs once, at home, before Session 1, following Lesson 0.1. Your instructor sends a setup guide and an environment check a week ahead.

✉️ FOR INSTRUCTORS

Want to teach this course? A complete instructor kit for Machine Learning Foundations (facilitator guides, slides, labs, workbooks, answer keys and assessments for all 11 sessions) is available under license. Ask about the instructor kit.

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